

Antonio Conopio

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EDUCATION

Bachelor of Computing, Software Engineering (Co-op), Minor in Math

Sep 2023 – Apr 2028

University of Guelph

Guelph, ON

- **Cumulative GPA:** 3.89 | Dean's List: 2023 - present

EXPERIENCE

Data Integration Intern – API Developer

Sep 2025 – Aug 2026

Co-operators

Guelph, ON

- Migrated the team's internal API template from R/Plumber to Python/FastAPI, reducing average response latency by 33%, improving modularity and input validation, and raising base test coverage from 40% to 97%.
- Designed and implemented new API features from stakeholder requirements, including business-layer logic and validation, with end-to-end testing and debugging across auto and home Risk Assist APIs.
- Extended auto and home Risk Assist APIs with new fields, validation logic, and business rules to onboard two new business channels, increasing API call volume by 30%.
- Led testing for the National Auto Risk Assist API, one of the team's flagship projects of the year, debugging across multiple models and data transformations to validate 100% parity with the modelers' pipeline.
- Built and deployed an R/Plumber API on Posit, with an automation script that migrates 1M+ API logs to Azure Data Lake Storage (ADLS), replacing manual extraction as part of the team's cloud modernization initiative.

PROJECTS

Orion | *React, TypeScript, Redux, Go, Python, PostgreSQL, Redis, WebSockets, Docker, Kubernetes*

- A distributed DAG workflow orchestration platform with a real-time dashboard for defining, scheduling, and monitoring job pipelines.
- Engineered a real-time WebSocket hub in Go that broadcasts live task status updates to connected clients, eliminating the need for polling.
- Designed a Redis Streams consumer group architecture for distributed Python workers with at-least-once delivery guarantees and automatic task retry on failure.
- Containerized all services with Docker and Kubernetes, enabling horizontal worker scaling based on Redis queue depth.

HomeHero | *SwiftUI, Spring Boot, Supabase, TCA*

- Developed a full-stack household management app for organizing tasks, shopping lists, expenses, and shared home data.
- Built a SwiftUI frontend using The Composable Architecture (TCA) with async/await to manage complex state, side effects, and predictable data flow.
- Implemented a Spring Boot REST API for authentication, task operations, and secure multi-user data synchronization.

HistoryHaven | *React, JavaScript, Python, FastAPI, scikit-learn, pandas, Vite*

- A full-stack analytics platform visualizing 19th-century Old Bailey court records, letting users filter, compare, and forecast historical crime trends by type, gender, and age group.
- Built a FastAPI backend from scratch, designing REST endpoints for filtered aggregation, cohort comparison, and summary statistics over a 130k+ row dataset spanning 1834–1913.
- Engineered a scikit-learn forecasting model with polynomial year features and log-transformed targets, using a time-based train/test split and leverage-based prediction intervals to generate confidence bounds under configurable economic scenarios.

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, JavaScript, Go, Swift, C, SQL, R, HTML/CSS

Frameworks: FastAPI, Spring Boot, React.js, Node.js, Express.js, SwiftUI, Redux, Tailwind CSS, Plumber

Databases: PostgreSQL, Redis, Supabase, Firebase

Cloud & DevOps: Docker, Kubernetes, Azure (ADLS), Posit Connect, Git, CI/CD

Libraries & Tools: pytest, pandas, NumPy, scikit-learn, Matplotlib, Databricks, Apache Spark